

NutShell L1 DCache 验证报告

开源 Python DV 环境 | 更新日期 2026-10-10 | SimpleBus 模块级验证

面向嵌入式与边缘 RISC-V 工作负载，构建可复现、无需商业仿真器的 Cache 验证流程。当前回归通过 39/39 个 pytest cases，命中 37/37 个功能覆盖目标。

DUT 与环境

项目	配置 / 工具
Cache	32-KiB, 4-way set-associative, 128 sets
Line / datapath	64-byte cache line, 64-bit datapath
开源工具链	Python, Toffee, Picker, Verilator, pytest, Yosys, SymbiYosys

验证架构

Sequences / directed + random
CPU, memory, MMIO, coherence agents + monitors
Independent memory + tag/valid/dirty/LFSR Refm
In-order scoreboard + protocol / ready-valid checkers
Functional coverage closure + Verilator line coverage

Tag model 独立预测 hit/miss、invalid-first refill way 和精确 LFSR victim way；scoreboard 核对 dirty victim 地址及 writeback 的八个 data beat。

回归与覆盖率

Metric	Result	Scope
pytest	39/39 pass	Directed + randomized regression
Functional coverage	37/37 (100%)	All planned closure goals hit
Maintained RTL line	98.0% (845/862)	Raw; 100% (845/845) after reviewed waivers
All-source line	95.1% (1423/1496)	Includes generated DUT/wrapper
Fault injection	4/4 caught	Representative tag, writeback, ready-valid, response faults

随机回归：本地 6 个固定 seed x 64 笔 = 384 笔操作；干净环境 CI 使用 8 个固定 seed = 512 笔操作。

关键场景

- 命中/缺失、partial byte write、same-word forwarding、跨 line 访问和 MMIO bypass
- set occupancy、4-way 冲突替换、clean/dirty victim、8-beat refill/writeback
- CPU 与 memory backpressure、排队 miss、coherence probe/release、reset 中止与恢复

形式验证与边界

make formal 执行三个 SymbiYosys gate, 并以 Yosys SAT 对 Stage3 做 8-cycle bounded safety check.

Gate	检查范围	边界 / 假设
Replacement selector	LFSR recurrence, nonzero reset state	SymbiYosys proof
Cache way selector	Replacement/refill one-hot selection	Unique-tag invariant
Cache Stage2	Request payload stable during backpressure	SymbiYosys proof
Cache Stage3	Response provenance/duplicate retirement, reset cancellation, stalled valid/command stability	8 cycles, data arrays abstracted; bounded only

解释与限制

- Stage3 的 bounded SAT 不是无界端到端响应生命周期证明; full-cache reset proof 仍是后续工作。
- 完整 response data-payload 稳定由仿真 checker 检查; formal harness 使用稳定至完成的单拍 READ/WRITE 及 MMIO 合约。
- 原始维护 RTL 覆盖率仍为 98.0%; waiver 调整值单独报告, waiver 不替代 raw coverage。
- 本项目是开源工具链下的模块级 sign-off-style regression, 不声称 commercial sign-off 或 SoC 级验证。

复现与 CI

```
cd workspace/nutshell_cache
source scripts/env.sh
make gen_dut
make formal
make report
```

GitHub Actions 在干净 Ubuntu runner 上安装开源工具链并运行上述 formal、8-seed 回归及覆盖率 gate。运行记录: github.com/cwu766485-ctrl/open_verif_learning/actions/workflows/nutshell-cache-dv.yml